Metan.iQ

Decision Intelligence Platform
for biomethane DM 2022 / RED III

USER MANUAL

Complete operating guide - features, calculations, workflow

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1. Software scope and regulatory context

Metan.iQ is a **Decision Intelligence Platform** designed for operators of **biomethane plants from anaerobic digestion with grid injection**, regulated by the Italian **Ministerial Decree 15 September 2022** (biomethane grid injection incentives) and the European directives **RED II (2018/2001)** and **RED III (2023/2413)**.

The software operates in **biomethane-only** mode (APP_MODE = 'biometano' hardcoded in the runtime). Any CHP self-consumption sections refer to **internal thermal/electrical recovery from self-consumption of a fraction of produced biomethane**, not to a standalone biogas-CHP plant injecting electricity into the grid.

1.1 Main objectives

- **Track** daily feedstock loaded and net Sm3 injected (REMI cabin readings).
- **Compute** GHG saving (%) per RED III Annex V Part C and UNI/TS 11567:2024.
- **Verify** the saving threshold on the MONTHLY sustainability lot.
- **Verify** the authorisation cap (e.g. 300 Sm3/h monthly average).
- **Optimise** the feedstock mix via LP solver to maximise GHG saving or tariff revenue.
- **Plan** the 15-year business plan with DM 2022 tariffs, matrix premiums, PNRR capital grants.
- **Export** audit-ready dossiers (PDF, Excel, CSV) for consultants, certification bodies, GSE.

1.2 Applied regulatory references

DM 15/09/2022	Scope of application
DM 15/09/2022	Biomethane grid tariffs (size-based, 15 years)
RED III (Dir. UE 2023/2413)	GHG saving thresholds, methodology Annex V Part C
D.Lgs. 199/2021	Italian transposition of RED II
UNI/TS 11567:2024	GHG emissions calculation for biofuels, standard yields
GSE Linee Guida 2024	Operating procedures for DM 2022 (Decree 248/2024)
JEC WTT v5	Joint Research Centre Well-to-Tank, default emission factors
Reg. UE 2022/996	Updated GWP values (CH_4 = 28, N_2O = 265)
UNI EN 16723-1	Biomethane specifications for grid injection (LHV, %CH_4)

1.3 Fundamental compliance rule

GHG sustainability is verified on the MONTHLY LOT, not on the individual day. Daily saving in the table is purely informative. The official **Compliant / Non Compliant** verdict is computed on the monthly aggregate, following the **Sustainability Lots (LS)** rule of UNI/TS 11567:2024 (max duration 6 months, monthly by GSE practice).

A month can be **Compliant** even when some isolated days fall below threshold, provided the energy-weighted monthly average respects the limit. This allows offsetting 'bad' days (pure maize) with 'good' days (manure, manure credit).

2. Architecture and technologies

Metan.iQ is a web application built on **Streamlit**. The user interacts through the browser, calculations run server-side in Python.

2.1 Technical stack

Frontend / runtime	Streamlit 1.57+ (Python 3.11+)
Calculation engine	NumPy, SciPy (LP solver), pandas
Visualisation	Plotly, HTML/CSS Material 3 custom
Export PDF	ReportLab (Metan.iQ brand template)
Export Excel	openpyxl (4-6 sheets, conditional formatting)
Export PPTX	python-pptx (8 yearly slides)
Local persistence	SQLite (data/metaniq_daily.db)
i18n	IT/EN dictionary substring replacement
Theme	Light / Dark switcher (navy + amber, Outfit font)
Hosting	Streamlit Cloud (private, branch dm2022-only)

3. Launching the app: first screen and sidebar

Opening the app URL presents a screen split into:

- **Left sidebar**: global selectors (language, regime, month, feedstocks, plant parameters).
- **Main area on the right**: three macro-tabs (Standard / Analysis / Results).
- **Top header**: Metan.iQ brand with version and active regime.
- **Footer**: legal links (Privacy, Terms), software version.

3.1 Sidebar: global control panel

The sidebar contains the following control groups:

Section	Control	Function
Language	Selector IT / EN	Switches labels, number and date format
Theme	Light / Dark toggle	WCAG AA palette switcher
Year / Month	Number input / Select	Working period
Plant ID	Text input	Multi-plant discriminator
Regime	Radio DM 2022 / DM 2018 / FER2	Changes GHG threshold and cap
Active feedstocks	Multiselect	Available columns in table
Plant net Sm3/h	Number input (default 300)	Authorisation cap
Aux factor	Auto or manual	Gross Sm3 / Net (default 1.29)
ep total	Auto or manual	Processing emissions (gCO2eq/MJ)
Manure credit	Checkbox (default ON)	Enables negative eec for manure

3.2 The four configuration modes

The main area presents the four tabs as a **numbered step selector (1 - 2 - 3 - 4)**, grouped into two families: **operational** (1-2, monthly operation on real data) and **strategic** (3-4, annual summary and economics), separated by a visual gap. Each mode is a card with a title and subtitle; the active card is highlighted in navy with a brass accent bar. On a narrow screen the cards stack vertically.

The four modes compared:

#	Mode	What it configures	When to use it
1	Standard	Standard values only (UNI-TS / RED III)	Routine daily operation, without lab analysis.
2	Analysis	Standard + custom values (BMT & EF)	When you have BMT lab yields or custom emission factors.
3	Annual Results	Real 12-month actuals (DB data)	See production, saving and charts on actually injected data.
4	Planning & BP	DM 2022 incentives, what-if scenario, 15-year pro forma	Configure tariffs, simulate and build the business plan.

Key distinction: mode 3 shows the **REAL actuals** (forward, from saved data), mode 4 works on a **planning scenario** (inverse solver). All share the month, feedstocks and sidebar parameters.

4. Tab 1 - Daily Operations Standard (UNI-TS / RED III)

Mode 1 - Standard (subtitle: *Standard values only - UNI-TS / RED III*). Main working tab for **daily management**. Uses **standard** yields and emission factors from UNI/TS 11567:2024 (Schedule A.5). Recommended for routine use without BMT analysis.

4.1 Daily feedstock table

Central table: one row per day of the month, dynamic columns based on the feedstocks selected in the sidebar.

Typical columns:

- **Date** - auto-generated.
- **Feedstock 1, 2, ...** - one column per active feedstock, input in t/day.
- **Gross Sm3** - computed (mass x yield).
- **Net Sm3** - **ENTERED BY OPERATOR** next day (REMI reading).
- **Net Sm3/h** - Net Sm3 / operating hours.
- **MWh** - Net Sm3 x NM3_TO_MWH (0.00979).
- **eec/esca/etd/ep** - component emissions.
- **e_total** - sum in gCO2eq/MJ.
- **Daily saving** - (80 - e_total) / 80 x 100, informative only.
- **Cap OK** - True if Net Sm3/h <= cap.
- **Cumulative Sm3/MWh/t** - month-to-date progressives.

4.3 Monthly KPIs

Below the table, 4 main metrics are displayed:

- **Monthly feedstock (t)**
- **Net Sm3 month**
- **Net MWh month**
- **GHG Saving (%)** with **COMPLIANT** / **NON COMPLIANT** badge

5. Tab 2 - Daily Operations Analysis (BMT / EF Override)

Mode 2 - Analysis (subtitle: *Standard + custom values - BMT & EF*). Advanced variant of Tab 1, for operators with BMT (Biological Methane Potential lab test) and/or Technical Report with custom emission factors. Custom values **override** the tabular ones feedstock by feedstock; with no override the mode computes identically to Standard.

5.1 BMT override (yields)

Replaces standard yields A.3 UNI/TS 11567:2024 with lab values. Mass balance reconciliation must stay within +/-15% for valid LS.

5.2 Emission factors override

- **eec** custom for non-tabulated feedstocks or supplier declarations.
- **ep** custom from real plant energy balance.
- **etd** custom from logistic survey.

6. Tab 3-4 - Annual Results and Planning & BP

Strategic family — two distinct tabs. **3 Annual Results**: REAL 12-month actuals (production, saving, charts on actually injected data). **4 Planning & Business Plan**: DM 2022 incentives, what-if scenario with inverse solver, 15-year pro forma. The sections below describe the contents, split across the two tabs.

6.1 Plant configuration (ep)

Defines the energy balance, which determines **aux_factor** (upgrading technology, off-gas handling, heat/electricity sources) and **ep_total** (sum of EP_UPGRADING + EP_OFFGAS + EP_HEAT + EP_ELEC). ep_total is constrained to **>= 0** per RED III Annex V Part C.

6.2 DM 2022 - Tariffs and premiums

- Plant size (Sm3/h bracket) determines the reference tariff (TR).
- Auction discount applied in GSE tender.
- Cumulative matrix premiums (Annex IX, recovered kWh, farm labour, cogen).
- PNRR capital grant (max 40% contribution).

6.3 15-year business plan

- CAPEX by line item, scaled with plant_size.
- Annual OPEX by line item (O&M;, manager, service, insurance).
- Leveraged financing (default 70% leverage, 5% rate, 15 years).
- Income statement: tariff revenue, OPEX, depreciation, taxes.
- Free Cash Flow + NPV, IRR, payback.

7. GHG calculation logic (RED III Annex V Part C)

This chapter analytically describes the GHG formula. It is the regulatory-technical core of the product: every output number derives from these relations.

7.1 RED III general formula

For each Sustainability Lot (LS = monthly aggregate):

E = eec + el + ep + etd + eu - esca - eccs - eccr
[gCO2eq / MJ biomethane]

Saving (%) = (FFC - E) / FFC x 100

Term	Meaning	Typical biomethane
eec	Feedstock extraction/cultivation	0 (waste) to +29 (maize)
el	Processing (digestion)	included in ep in this model
ep	Processing (upgrading, heat, electr.)	0-10, cap >= 0
etd	Transport and distribution	0.8 (standard default)
eu	Fuel use	0 (grid biomethane)
esca	Soil C accumulation credit	0 (default)
eccs/eccr	CO2 capture credit	0 (default)
FFC	Fossil fuel comparator	80 grid/heat, 94 transport, 183 electricity

7.2 Aggregation on the monthly lot

It is NOT an arithmetic mean of daily savings. Per UNI/TS 11567:2024 and RED III Annex V Part C, aggregation is energy-weighted on the total lot:

e_w_lot = SUM (e_feedstock_i x Energy_feedstock_i) / SUM (Energy_feedstock_i)

Energy_i = mass_i x CH4_yield_i x LHV
e_feedstock_i = eec_i + esca_i + etd_i + ep

Saving_lot = (FFC - e_w_lot) / FFC x 100

Days with daily saving <80% do not make the month non-compliant: they are diluted by high-saving days through energy weighting.

7.3 eec defensibility tiers (A/B/C/D)

Tier	Meaning	Typical eec
A - Regulatory default	Tabulated Schedule A.5 UNI/TS	Maize 29, Sorghum 26, waste 0
B - Zero by rule	Non-tabulated Annex IX residue	0 (residue rule)
C - Conservative estimate	JEC/KTBL literature against operator	positive, non-contestable
D - Manure credit	Animal manure credit	negative (-45 std), supplier baseline required

7.4 Manure credit (Tier D)

For animal manure the software applies by default eec **-45 gCO2eq/MJ** (RED III Annex V/VI). It reflects methane emissions **avoided** compared to open tank/lagoon storage.

The manure credit requires a **supplier baseline declaration** during certification body audit. If the supplier uses a COVERED tank with capture or N-stripping, the credit must be reduced or removed.

8. aux_factor calculation and energy balance

The **aux_factor** is the ratio of gross biomethane Sm3 to net Sm3 injected. It accounts for the plant's energy self-consumption.

8.1 Formula

aux_factor = 1 / (1 - f_heat - f_elec - f_slip - f_margin)

Net_Sm3 = Gross_Sm3 / aux_factor

- **f_heat** - biomethane fraction for digester boiler (0 if external heat).
- **f_elec** - fraction burned in internal CHP (0 if grid or PV).
- **f_slip** - CH4 losses in upgrading (0.5%-2%).
- **f_margin** - prudential margin (3% default).

8.2 Default and typical ranges

DEFAULT_AUX_FACTOR	1.29 (JRC-CONCAWE)
Amine + thermal self-consumption	1.20 - 1.32
PSA + grid electricity	1.10 - 1.18
Membranes + cogen	1.15 - 1.25
Manual override	Available in 'Config Tecnica & GHG'

9. Regulatory constraints and monthly compliance check

The software automatically verifies TWO month-end constraints:

9.1 Constraint A - GHG saving threshold

Plant commissioning date	End use	Threshold
Before 5 Oct 2015	Transport	50%
5 Oct 2015 - 31 Dec 2020	Transport	60%
From 1 Jan 2021	Transport	65%
From 1 Jan 2021	Electricity / heat (>1 MW)	70%
From 1 Jan 2026 (new plants)	Electricity / heat	80%

The relevant date is the **commissioning date of the ORIGINAL biogas plant**, NOT of the upgrading.

9.2 Constraint B - Authorisation cap Sm3/h

Verifies that the monthly average flow rate does not exceed the AUA / AIA cap. Default 300 Sm3/h, configurable in the sidebar.

9.3 Final outcome

The month is **COMPLIANT** if BOTH constraints are met. The monthly PDF shows a green/red badge and margin in percentage points.

10. Export: PDF, Excel, CSV

10.1 Monthly PDF (7 pages)

Page	Content
1	Cover - Official outcome, main KPIs
2	Monthly KPI summary + regulatory constraints
3	Operational guidance (month-end corrective actions)
4-5	Detailed daily table (26 columns)
6	Simulation registry & Audit Trail
7	Regulatory references & Validation

10.2 Excel (4 sheets)

- **Monthly Summary** - outcome, main KPIs, feedstock totals.
- **Daily Operations** - 26-column table, 28-31 rows.
- **Registry & Parameters** - traceability entries for audit.
- **Constraints** - outcome for each regime constraint.

10.3 CSV

Separator **;**, decimal **,**, UTF-8 with BOM. Opens directly in Italian Excel. Contains the complete daily table.

10.4 Certification Body Conformity Dossier

- All calculation parameters with regulatory source.
- Lot mass balance: feedstock vs biomethane deviation (<= +/- 15%).
- Yield origin (standard / BMT).
- Emission factors origin (standard / Technical Report).
- Incoming DDTs and supplier list.
- Supplier sustainability declarations.

11. Plant registry and audit trail

Mandatory section for GSE / certification body traceability. All values appear on the cover of reports and in the Excel.

Item	Notes
Company name	Full registered name
Registered office	Registered office address
Plant name	Operating name
Operating site	Plant address
Applied regime	DM 2022 / DM 2018 / FER2
Regulatory threshold (%)	80 / 70 / 65 / 60 / 50 depending on regime
Fossil comparator	80 / 94 / 183 based on end use
Aux factor	From Tech Config or override
EP total	From Tech Config or override
Plant net (Sm3/h)	Authorisation cap
Yield origin	BMT override if active, else standard
Emission factors origin	Tech Report override if active, else UNI-TS
Days with data	Count of filled days
Days cap violated	Count of days with Sm3/h > cap

12. FAQ and troubleshooting

12.1 Frequently asked questions

Q: Why is monthly saving not the average of daily savings?

A: Because UNI/TS 11567:2024 and RED III mandate aggregation on the lot energy. See ch. 7.2.

Q: Can I have days with saving < 80% without being Non Compliant?

A: Yes, provided the aggregate monthly saving is >= 80%. Offsetting high/low days is a feature of LS.

Q: What if I don't enter the net Sm3 REMI?

A: Flow, net MWh and revenue KPIs are shown as 0. GHG saving can still be computed from feedstock.

Q: Does the manure credit -45 always apply?

A: Yes by default, only if the supplier stores manure in an OPEN tank. Covered tank + capture cancels the credit.

Q: Can ep_total be negative?

A: No. Per RED III Annex V Part C, ep >= 0. The software applies an automatic floor.

Q: Which biomethane LHV does the software use?

A: 9.79 kWh/Sm3 = 35.24 MJ/Sm3, compliant with UNI EN 16723-1 (grid spec biomethane, ~98% CH4).

Q: Is data persistent on Streamlit Cloud?

A: The Streamlit Cloud filesystem is EPHEMERAL. Export PDF/Excel regularly as backup.

Q: How do I handle a non-certified supplier?

A: It must be excluded from the LS or a certification process must be started.

12.2 Troubleshooting

Problem	Probable cause / Action
All KPIs at 0 after opening	Sidebar: no feedstock selected. Add at least one.
Unexpectedly low monthly saving	Too much maize (eec +29). Increase manure.
Net MWh != daily cumulative	Upgrade to v0.5.0+ (UNI EN 16723-1 fix).
ep_total shows 0 but config says otherwise	Floor ep >= 0: config would produce negative ep (RTO off-gas).
Table not editable	Month already closed. Use 'New month' or reopen.
LP solver no solution	Incompatible constraints. Relax one of the three.

Appendix A - Complete regulatory formulas

A.1 GHG per single feedstock

```
e_feedstock = eec + esca + etd + ep

All in gCO2eq/MJ. Manure credit on eec with negative sign (e.g. -45 for slurry).
```

A.2 Feedstock energy

```
Energy_i [MJ] = mass_i [t] x CH4_yield_i [Nm3/t] x LHV [MJ/Nm3]

LHV = 35.24 MJ/Nm3 (UNI EN 16723-1)
```

A.3 Lot aggregate

```
e_w_lot = SUM(e_feedstock_i x Energy_i) / SUM(Energy_i)
Saving = (FFC - e_w_lot) / FFC x 100
FFC = 80 (grid/heat) | 94 (transport) | 183 (electricity)
```

A.4 LS mass balance

```
UNI/TS 11567 sec. 6 tolerance:
| expected_biomethane - REMI_biomethane | / expected_biomethane <= 15%
Maximum LS duration: 6 months (GSE practice: monthly).
```

Appendix B - Physical constants and emission factors

Constant	Value	Source
Biomethane LHV	35.24 MJ/Nm3 = 9.79 kWh/Sm3	UNI EN 16723-1
NM3_TO_MWH	0.00979	LHV-derived
Default CH4 purity	98.0 %	UNI EN 16723-1
FFC grid / heat	80 gCO2eq/MJ	RED III All. VI
FFC transport	94 gCO2eq/MJ	RED III All. VI
FFC electricity	183 gCO2eq/MJ	RED III All. VI
GWP CH4	28	Reg. UE 2022/996
GWP N2O	265	Reg. UE 2022/996
GWP CO2	1	Reg. UE 2022/996
DEFAULT_AUX_FACTOR	1.29	JRC-CONCAWE
GHG threshold new grid/heat 2026+	80 %	RED III
GHG threshold transport 2021+	65 %	RED III
LS mass balance tolerance	+/- 15 %	UNI/TS 11567:2024 sec.6
Maximum LS duration	6 months	UNI/TS 11567:2024 sec.6
Document retention	5 years	UNI/TS 11567:2024

Appendix C - Complete feedstock database (42 entries)

Complete list of the **42 feedstocks** in the software's FEEDSTOCK_DB, extracted directly from *app_mensile.py*. Regulatory values from UNI/TS 11567:2024 Schedule A.5 unless overridden by BMT / Technical Report. Feedstock names are kept in Italian as they are the official commercial designations used in Italian audits (RINA / SGS / ISCC) and GSE filings.

#	Feedstock (IT)	Category	eec	etd	Yield Nm3CH4/t	Tier
1	Trinciato di mais	Energy crops	+29	0.8	116.1	A
2	Trinciato di sorgo da foraggio	Energy crops	+26	0.8	81.4	A
3	Trinciato di sorgo (energetico)	Energy crops	+26	0.8	90.5	A
4	Triticale insilato	Energy crops	+20	0.8	106.1	A
5	Segale insilata	Energy crops	+22	0.8	80.0	A
6	Orzo insilato	Energy crops	+22	0.8	82.0	A
7	Loietto insilato (ryegrass)	Energy crops	+18	0.8	93.7	A
8	Erba medica insilata	Energy crops	+15	0.8	70.0	A
9	Doppia coltura (2° raccolto)	Energy crops	+15	0.8	95.0	A
10	Erbaio misto insilato	Energy crops	+16	0.8	64.0	A
11	Barbabietola da zucchero	Energy crops	+12	0.8	105.0	A
12	Liquame suino	Animal manure	-45	0.8	14.0	D
13	Liquame bovino	Animal manure	-45	0.8	14.0	D
14	Liquame bufalino	Animal manure	-45	0.8	14.0	D
15	Letame bovino palabile	Animal manure	-30	0.8	35.0	D
16	Letame equino	Animal manure	-20	0.8	42.0	D
17	Pollina ovaiole (aerobico)	Animal manure	+5	0.8	84.0	C
18	Pollina broiler (lettiera)	Animal manure	-15	0.8	105.0	D
19	Pollina tacchini	Animal manure	-10	0.8	100.0	D
20	Deiezioni conigli	Animal manure	+5	0.8	75.0	C
21	Sansa di olive umida	Agro-industrial by-products	+3	0.8	120.0	C
22	Sansa vergine	Agro-industrial by-products	+2	0.8	140.0	C
23	Pastazzo di agrumi	Agro-industrial by-products	+6	0.8	100.0	C
24	Vinaccia (con raspi)	Agro-industrial by-products	+5	0.8	130.0	C
25	Raspi d'uva	Agro-industrial by-products	+3	0.8	70.0	C
26	Feccia vinicola	Agro-industrial by-products	+3	0.8	180.0	C
27	Siero di latte	Agro-industrial by-products	+3	0.8	30.0	C
28	Scotta (siero residuo)	Agro-industrial by-products	+2	0.8	22.0	C
29	Trebbie di birra	Agro-industrial by-products	+4	0.8	140.0	C

#	Feedstock (IT)	Category	eec	etd	Yield Nm3CH4/t	Tier
30	Lolla/pula di riso	Agro-industrial by-products	+2	0.8	50.0	C
31	Melasso	Agro-industrial by-products	+8	0.8	180.0	A
32	Scarti panificazione/pasticceria	Agro-industrial by-products	+5	0.8	280.0	C
33	Grassi esausti / UCO	Agro-industrial by-products	+2	0.8	700.0	C
34	Scarti macellazione (cat. 3)	Agro-industrial by-products	+5	0.8	180.0	C
35	Sottoprodotti ortofrutticoli	Agro-industrial by-products	+7	0.8	100.0	A
36	Scarti caseari vari	Agro-industrial by-products	+4	0.8	40.0	C
37	Fanghi agro-industriali	Agro-industrial by-products	+3	0.8	55.0	C
38	Polpe di barbabietola fresche	Agro-industrial by-products	0	2.0	50.0	A/B
39	Polpe di barbabietola insilate	Agro-industrial by-products	0	2.5	75.0	A/B
40	Melasso di barbabietola	Agro-industrial by-products	0	1.5	280.0	A/B
41	FORSU selezionata	OFMSW / Waste	0	0.8	64.8	A/B
42	Fanghi depurazione	OFMSW / Waste	0	0.8	13.0	A/B

Tier legend (eec defensibility level in certification body audit):
A = Tabulated regulatory default (Schedule A.5 UNI/TS 11567:2024).
A/B = eec = 0 by rule (waste, by-products, residues).
C = Conservative positive estimate (no manure credit). Against operator interest, non-contestable in audit.
D = Manure credit (negative eec). Requires **supplier baseline declaration** (open tank/lagoon storage).

Distribution of the 42 entries: Energy crops 11 | Animal manure 9 | Agro-industrial by-products 20 | OFMSW/Waste 2.

End of manual. Support: carlo.sicurini@gmail.com.